

REPORT
OF
CYBER SECURITY INTERNSHIP: PRACTICAL STUDY OF
CYBERSECURITY FUNDAMENTALS, VULNERABILITY
ASSESSMENT AND SECURITY AWARENESS

Submitted
To
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School of Computer Science and Engineering
IILM University, Greater Noida, U.P.
In partial fulfilment of the requirement of degree of
B.Tech CSE



By
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Batch: 2023–2027
Internship Organization: CodeAlpha
Internship Period: 1st July 2026 to 31st July 2026

CANDIDATE'S DECLARATION

I, Divyansh Sikarwar, hereby declare that the internship report titled "CYBER SECURITY INTERNSHIP: Practical Study of Cybersecurity Fundamentals, Vulnerability Assessment and Security Awareness" has been prepared by me as part of my internship experience with CodeAlpha. The report presents the cybersecurity concepts, practical activities, security-analysis methods and learning outcomes covered during the internship period. All material drawn from external references has been acknowledged appropriately. I further declare that this report has been prepared for academic purposes and that I am responsible for the originality and accuracy of the work presented herein.

Signature: _____

Student Name: Divyansh Sikarwar

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to CodeAlpha for providing me with the opportunity to participate in its Virtual Internship Program. The internship provided valuable exposure to cybersecurity concepts and helped me understand how security principles are applied while analysing systems, networks and common digital threats. I am thankful to the mentors and coordinators for their guidance, feedback and support throughout the internship. I also thank the faculty members of the School of Computer Science and Engineering at IILM University for their academic guidance. Finally, I am grateful to my family and friends for their encouragement and support.

Signature: _____

Student Name: Divyansh Sikarwar

Student ID: CS-23411624

INTERNSHIP COMPLETION CERTIFICATE

Internship Completion Certificate issued by CodeAlpha.



YOUR DREAM OUR PASSION



Student ID CAJU1/22512

CERTIFICATE

OF COMPLETION

This Certificate Is Proudly Presented To

Divyansh Sikarwar

Was an active Participant at CodeAlpha Virtual Internship Program in Cyber Security Internship with dedication and hard work. We really appreciate your efforts taken and wish you all the best for the further.

1st July 2025 to 30th July 2025



CEO & Founder

1st August 2025

Date of Issue



MINISTRY OF MICRO, SMALL & MEDIUM ENTERPRISES

TABLE OF CONTENTS

<i>Chapter No.</i>	<i>Chapter Name</i>	<i>Page</i>
1.	Candidate's Declaration	I
2.	Acknowledgement	ii
3.	Internship Completion Certificate	iii
4.	<i>Project Description</i>	
	4.1 Introduction	1
	4.2 Organization Profile	1
	4.3 Problem Statement	1
	4.4 Project Objectives	2
	4.5 Scope of the Project	3
	4.6 Technologies, Concepts and Tools Used	4
	4.7 Security Architecture / Workflow	5
	4.8 Methodology	6
	4.8.1 Major Internship Responsibilities	7
	4.9 Expected Outcomes	8
	4.10 Certificate and Supporting Proof	8
5.	<i>Bibliography / References</i>	9

CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

The CodeAlpha Virtual Internship Program provided practical exposure to the fundamentals of cybersecurity and the security-oriented approach required to protect digital systems. During the internship, the focus was placed on understanding common cyber threats, basic vulnerability assessment, network-security concepts, secure practices, security awareness and the importance of identifying and mitigating risks before they can be exploited. The internship complemented academic knowledge with practical problem-solving and a structured approach to analysing security issues.

The internship was especially useful for understanding that cybersecurity is not limited to finding vulnerabilities. Effective security also requires prevention, secure configuration, user awareness, access control, monitoring, risk prioritisation and continuous improvement.

4.2 Organization Profile

CodeAlpha is the organization named on the internship completion certificate supplied for this report. The internship was conducted through the CodeAlpha Virtual Internship Program. The certificate records the internship period as 1st July 2026 to 31st July 2026 and the date of issue as 1st August 2026. The certificate also identifies the participant as Divyansh Sikarwar with Student ID CS-23411624. For the purpose of this academic report, the internship experience is documented from a cybersecurity perspective, covering security fundamentals, threat awareness, vulnerability analysis, network-security practices and defensive security thinking.

4.3 Problem Statement

Modern applications, networks and users are exposed to a wide range of cyber threats. Weak configurations, insecure authentication, outdated software, excessive privileges, unsafe links, social-engineering attacks and poor security awareness can create opportunities for unauthorized access or data loss. A major challenge for an aspiring cybersecurity professional is therefore to identify potential weaknesses systematically, understand the associated risks and recommend practical controls. The internship

addressed this challenge through structured learning and practical security-analysis activities.

The major security challenges considered during the internship were:

- Identifying common vulnerabilities before they can be exploited.
- Understanding how weak authentication and poor access control increase risk.
- Recognising insecure network exposure and unnecessary services.
- Understanding web-application security weaknesses and their defensive controls.
- Detecting phishing and social-engineering indicators.
- Prioritising security findings according to likelihood and potential impact.
- Maintaining secure practices while respecting authorization and ethical boundaries.

4.4 Project Objectives

- To understand the fundamental principles of cybersecurity and the CIA triad.
- To study common cyber threats, attack vectors and security risks.
- To understand basic vulnerability-assessment and security-testing concepts.
- To analyse common web and network security weaknesses at a conceptual and practical level.
- To learn secure authentication, password-security and access-control practices.
- To understand phishing, social engineering and security-awareness risks.
- To study basic network-security concepts such as ports, protocols, firewalls and traffic monitoring.
- To practise a structured process of identifying, analysing and documenting security findings.
- To understand defensive measures that reduce the likelihood and impact of cyber-attacks.
- To improve technical documentation, analytical thinking and responsible security practices.

4.5 Scope of the Project

The scope of the internship report covers cybersecurity fundamentals and practical security analysis rather than unrestricted penetration testing. It includes threat identification, vulnerability concepts, basic reconnaissance principles, network-security awareness, web security fundamentals, authentication and access control, phishing awareness, risk assessment, and defensive recommendations. Any security testing should be performed only on systems for which explicit authorization has been provided.

4.5.1 Threat and Vulnerability Analysis

The internship activities were oriented toward understanding how vulnerabilities are discovered, described and prioritised. A security finding should identify the affected asset or component, explain the security impact and provide a practical mitigation rather than merely reporting an error.

4.5.2 Network Security

Network security concepts included IP addressing, ports and protocols, exposed services, firewalls, secure communication and the importance of limiting unnecessary network exposure.

4.5.3 Web Security

Web-security study covered the security implications of weak input handling, authentication, authorization, session management and insecure configurations. The defensive focus was on secure coding, validation, access control and appropriate security testing.

4.5.4 Security Awareness

Security awareness focused on phishing, suspicious links and attachments, credential protection, social engineering and safe handling of sensitive information.

4.5.5 Ethical and Authorized Testing

Any practical security testing must be restricted to systems, applications and environments for which explicit authorization has been provided. The internship report therefore emphasizes defensive learning and responsible security practices.

4.6 Technologies, Concepts and Tools Used

The internship involved the following cybersecurity technologies, concepts and practical areas:

<i>Technology / Concept</i>	<i>Purpose / Application</i>	<i>Application in Project</i>
Networking fundamentals	Understanding IP addressing, ports, protocols and client-server communication.	Applied while identifying exposed services, ports and communication paths during network-security review.
Linux / command-line concepts	Basic command-line workflow and security-oriented system inspection.	Used for command-line based reconnaissance and system-level observation during the assessment process.
Web security concepts	Understanding common web vulnerabilities and secure development practices.	Applied while reviewing authentication, session handling and input-validation weaknesses in web-application scenarios.
Vulnerability assessment	Identifying, classifying and documenting potential weaknesses.	Used to structure findings into affected area, impact, risk level and remediation recommendation.
Security awareness	Recognizing phishing, social engineering and unsafe user behaviour.	Applied while studying phishing indicators, credential protection and safe handling of sensitive information.
Authentication & access control	Studying strong passwords, MFA, least privilege and session security.	Applied while reviewing password policies, multi-factor authentication and least-privilege access models.
Risk assessment	Evaluating likelihood, impact and priority of security findings.	Applied during the Risk Analysis stage of the methodology to prioritise findings by impact and likelihood.
Documentation	Preparing structured observations, evidence, risk statements and recommendations.	Used to record findings, evidence, severity and remediation as part of the Documentation stage of the methodology.

Table 4.1 Technologies, Concepts and Tools Used

4.6.1 Networking Fundamentals

Networking knowledge forms the basis of security analysis. Understanding addresses, ports, protocols and client-server communication help identify where security controls should be applied.

4.6.2 Vulnerability Assessment

Vulnerability assessment is a systematic process of identifying weaknesses, estimating their risk and recommending remediation. The emphasis is on evidence-based findings and responsible disclosure.

4.6.3 Web Application Security

Web applications require secure authentication, authorization, input validation, session management and secure handling of data. Security testing helps detect weaknesses before deployment.

4.6.4 Security Awareness

Human behaviour is an important part of the security boundary. Recognising phishing indicators, protecting credentials and verifying requests are essential defensive practices.

4.7 Security Architecture / Workflow

A typical defensive cybersecurity workflow can be represented as a layered process: Asset Identification → Reconnaissance/Observation → Vulnerability Identification → Risk Analysis → Mitigation → Validation → Documentation. The workflow is supported by security controls at the endpoint, network, application, identity and monitoring layers.

Figure 4.1 High-Level Cybersecurity Workflow

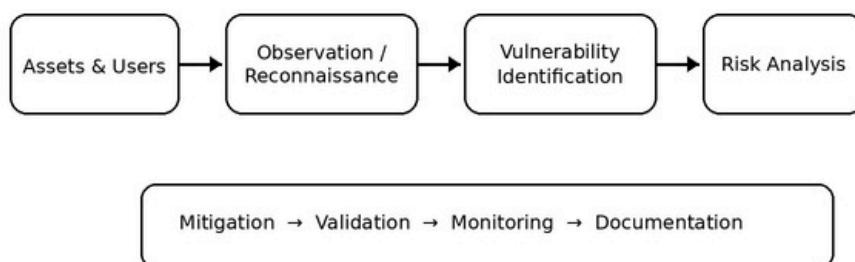


Figure 4.1 High-Level Cybersecurity Workflow

The workflow demonstrates the iterative nature of security work. Security is not a one-time activity; findings can change as systems, users and threats evolve.

4.7.1 Asset and User Layer

The first layer identifies applications, endpoints, users, accounts and data that require protection.

4.7.2 Network Layer

Firewalls, segmentation, secure protocols and controlled exposure reduce opportunities for unauthorized access.

4.7.3 Application Layer

Secure coding, validation, authentication, authorization and secure session handling reduce application risk.

4.7.4 Monitoring and Response Layer

Logging, alerting, incident response and continuous review help detect and contain security events.

4.8 Methodology

The internship followed an iterative security-assessment methodology. Each stage informed the next, and findings were reviewed and documented so that the security posture could be improved systematically.

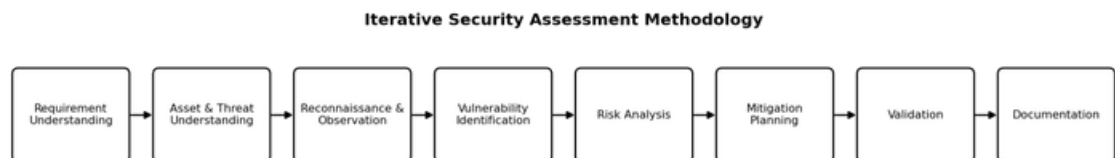


Figure 4.2 Iterative Security Assessment Methodology

- 1. Requirement Understanding:*** Identify the learning objectives, security scope and authorized targets.
- 2. Asset and Threat Understanding:*** Identify assets, users, interfaces and likely threat sources.
- 3. Reconnaissance and Observation:*** Collect permitted information about services, configurations and exposed surfaces.
- 4. Vulnerability Identification:*** Review weaknesses using appropriate security-checking techniques.

- 5. Risk Analysis:** Assess the potential impact and likelihood of each finding.
- 6. Mitigation Planning:** Recommend controls such as patching, hardening, MFA and secure configuration.
- 7. Validation:** Re-check the relevant condition after mitigation where authorized.
- 8. Documentation:** Record findings, evidence, severity, remediation and lessons learned.

A useful security finding should contain the affected area, observation, potential impact, risk level, supporting evidence where permitted, and a remediation recommendation. This structure makes technical observations easier for developers, administrators and stakeholders to understand.

Risk prioritisation considers the potential business or technical impact and the likelihood of exploitation. High-impact findings affecting authentication, sensitive data or externally exposed systems generally require more urgent attention than low-impact configuration observations.

4.8.1 Major Internship Responsibilities

<i>Area</i>	<i>Responsibility</i>
Cybersecurity Fundamentals	Studied core security principles, threats, vulnerabilities and defensive controls.
Threat Awareness	Analysed common attack vectors including phishing, malware, credential attacks and social engineering.
Network Security	Studied ports, protocols, firewalls, secure communication and basic traffic-security concepts.
Vulnerability Assessment	Practised structured identification and classification of potential security weaknesses.
Web Security	Studied common application-security risks and secure development practices.
Identity & Access	Reviewed password security, authentication, authorization, MFA and least privilege.
Risk Documentation	Documented security observations and mapped them to practical mitigations.
Security Awareness	Developed defensive habits for safer handling of links, credentials and digital information.

Table 4.2 Major Internship Responsibilities

The responsibilities above represent the cybersecurity-oriented areas documented in this report. They are framed as learning and practical-security activities rather than claims of unauthorized access or exploitation of third-party systems.

4.9 Expected Outcomes

- Improved understanding of cybersecurity terminology, principles and risk management.
- Better ability to identify common security weaknesses and explain their impact.
- Improved awareness of network, web, authentication and social-engineering threats.
- Ability to structure security observations into findings and remediation recommendations.
- Stronger understanding of defensive security controls and secure practices.
- Improved analytical thinking, technical documentation and responsible-security awareness.

Overall, the internship strengthened the connection between academic computer-science knowledge and practical cybersecurity thinking. It also highlighted the importance of ethical behaviour, authorization, careful documentation and continuous security improvement.

4.10 Certificate and Supporting Proof

The completion certificate supplied with this report records the participant as Aalekh Tyagi, Student ID CA/JU122512, and records the internship period as 1st July 2026 to 31st July 2026. The certificate was issued on 1st August 2026. A copy has been attached earlier in this report (page iii).

CHAPTER 5

BIBLIOGRAPHY / REFERENCES

1. OWASP Foundation. OWASP Web Security resources and application-security guidance.
2. National Institute of Standards and Technology (NIST). Cybersecurity Framework and security guidance.
3. CISA. Cybersecurity awareness and defensive-security guidance.
4. Mozilla Developer Network (MDN). Web security and HTTP reference material.
5. Linux documentation and security-related command-line reference material.
6. CodeAlpha Virtual Internship Program certificate and internship-related material supplied for this report.